

Supplementary Information
Beyond Arbitrariness: Detecting Geographic Structure
— *and the Limits of Current Linguistic Databases*

Author list omitted for anonymised review

Abstract

This Supplementary Information is an audit and reproducibility layer for the main manuscript. Every publication-facing quantity is traceable from the manuscript to an SI table or figure, from that table or figure to a Result CSV, and from the CSV to the exact notebook or Python code that generated it. All numerical values are derived programmatically from the Result CSV files; no value is transcribed from the manuscript, notebook display output, or PDF. The reported models are Beta-Binomial hierarchical regressions; the earlier Binomial fits are retained only as a superseded sensitivity layer. The expensive H100/GPU posterior models were *not* rerun to build this SI. Throughout, we distinguish word-final vowel occurrence (measured directly) from open-syllable structure (a plausible typological correlate), and treat latitude as a composite geographic coordinate rather than a causal exposure.

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S1. Data provenance and preprocessing

The two independently assembled databases have partially overlapping language coverage. Loan forms are excluded (`ASJP Loan == "true"`; Lexibank `Loan ∈ {true,1}`); languages require valid coordinates and a family identifier and must meet a minimum of 20 concepts. Word-initial and word-final vowel occurrence are extracted per form and aggregated to the language level. Table S1 summarises provenance; Table S2 gives the segment and vowel classification rules, taken verbatim from the notebook code.

Table S1: Database provenance and preprocessing summary.

Attribute	ASJP	Lexibank
Version / commit SHA	ASJP (CLDF), SHA 012795349540	Lexibank-analysed, SHA edf5f028de7e
Raw forms (pre-exclusion)	not recorded in committed CSVs	not recorded in committed CSVs
Retained word tokens	560,372	1,740,092
Pre-filter language summaries	11,393	5,501
Inferential languages	10,886	5,501
Family identifiers (primary)	646	298
Transcription system	ASJPcode (41 symbols)	IPA segments (CLTS)
Coordinate source	ASJP languages.csv Lat/Long	Lexibank CLDF Lat/Long
Family source	ASJP CLDF classification	Glottolog Family (→ Family_in_Data)

Note. Retained word tokens are summed from the family-count CSVs; raw pre-exclusion form totals are not stored in the committed Result CSVs. Family-identifier counts are the primary inferential sample (isolate/unknown languages carry unique identifiers).

Table S2: Segment and vowel classification rules.

Rule	ASJP	Lexibank (IPA)
Vowel definition	character in ASJP_VOWELS = {i e E 3 a u o}	token whose first non-diacritic character is an IPA vowel (25-symbol set)
Ignored / stripped symbols	keep only ASJP vowel/consonant symbols; all modifiers dropped	strip length, nasalisation, phonation, stress, tone diacritics (IPA_STRIP) before classifying
Segment extraction rule	per-character: retain symbols in the ASJP vowel or consonant set	split Segments on spaces (+→space); drop empty tokens
Initial-segment rule	initial vowel = 1 iff first segment is a vowel	initial vowel = 1 iff first segment is a vowel
Final-segment rule	final vowel = 1 iff last segment is a vowel; final cluster = trailing consonant count	same rule applied to IPA segments
Malformed / empty forms	non-string or no valid segment → empty list, language contributes no counts	non-string/empty or all-diacritic → dropped

Note. Definitions taken verbatim from notebook cell 6 (ASJP_VOWELS, IPA_VOWELS, IPA_STRIP, asjp_segments, ipa_segments, final_cluster) and the aggregation in cell 7. Word-final vowel occurrence is measured directly; it is a plausible correlate of, but not identical to, open-syllable structure.

S2. Sample flow and metadata audits

Table S3 traces the ASJP sample from the full language summary to the primary inferential sample and the macroarea-linked subset. 507 languages are removed by the 20-concept inclusion threshold. A further 260 languages could not be linked under the identifier used for the macroarea table; these are not missing-macroarea cases — all 260 are isolate/unknown languages carrying unique family identifiers. Table S4 gives the macroarea composition and Table S5 the linkage audit summary; the complete 260-language ledger is retained in the repository as `asjp_macroarea_dropped_languages.csv`.

Table S3: ASJP sample flow.

Stage	Languages	Family ids	Isolate/unknown
ASJP language summary before primary filter	11,393	667	278
Primary complete-case sample (n_concepts >= 20)	10,886	646	260
Macroarea complete-case sample	10,626	386	0

Note. 507 languages were removed by the 20-concept inclusion threshold (11,393 → 10,886). A further 260 languages could not be linked under the identifier used for the macroarea table; these are NOT missing-macroarea cases – all 260 are isolate/unknown languages with unique family identifiers (full ledger: `asjp_macroarea_dropped_languages.csv`).

Table S4: Macroarea-linked sample composition.

Macroarea	Languages	Family ids	Share of matched sample
Africa	3,570	53	33.6%
Papunesia	2,895	126	27.2%
Eurasia	2,655	38	25.0%
North America	692	72	6.5%
South America	508	86	4.8%
Australia	306	29	2.9%
Total	10,626	386	100.0%

Note. Matched macroarea-complete sample (10,626 languages, 386 family identifiers). Shares are of the matched sample; family identifiers are not additive across macroareas.

Table S5: Macroarea linkage audit summary.

Linkage outcome	Languages	Family ids	Isolate/unknown
included	10,626	386	0
no_macroarea_match_by_Language_ID	260	260	260

Note. Macroarea source: ASJP `languages.csv` column `Macroarea`; inner merge on `Language_ID`; 11,540 source rows, 0 duplicate identifiers, 11,115 languages with a macroarea. The complete 260-language exclusion ledger is retained as `asjp_macroarea_dropped_languages.csv`; every excluded language is isolate/unknown, not a missing-macroarea case.

S3. Statistical model specifications

The reported models are hierarchical Beta-Binomial regressions of the language-level count of word-final vowels out of total forms. The linear predictor is $\eta = \alpha + \beta_{\text{lat}} z_{\text{lat}} + \sigma_{\text{family}} z_{\text{family}[i]}$, with a Beta-Binomial observation model $y \sim \text{BetaBinomial}(n, \mu\phi, (1 - \mu)\phi)$, $\mu = \text{logit}^{-1}\eta$. Family random intercepts condition on family-level clustering; they are not full phylogenetic control. The positional model adds language-level intercepts, and the initial–final contrast is a within-language comparison. Latitude is the absolute value of the coordinate, z-scored with population standard deviation; it is a composite geographic coordinate, not a causal exposure. Priors are read directly from the code (Table S6), not inferred from posteriors. The full model inventory spans the primary model, the Lexibank replication, the joint initial–final positional model, the within/between-family decomposition, the matched-sample family-only and family-plus-macroarea models, the hemisphere-interaction and Southern-Hemisphere macroarea-specific models, the leave-family-out and leave-top-two positional re-fits, and the superseded Binomial sensitivity layer.

Table S6: Prior specification and parameterisation.

Parameter	Prior	Role
α (intercept)	Normal(0, 1.5)	global log-odds intercept
β_{lat} and all slope terms	Normal(0, 1)	latitude / level / interaction slopes (beta_lat, beta_lat_north, beta_south_level, beta_lat_south_delta, beta_between, beta_within, beta_lat_initial, beta_final_level, beta_lat_final_delta)
$\sigma_{\text{family}}, \sigma_{\text{language}}$	Exponential(1)	random-effect standard deviations (family; and language in the positional model)
$z_{\text{family}}, z_{\text{language}}$	Normal(0, 1), then mean-centred	non-centred, sum-to-zero standardised random intercepts
macroarea effects	Normal(0, 0.75)	fixed categorical intercepts/slopes for the six observed macroareas (deliberately NOT hierarchical)
ϕ (Beta-Binomial concentration)	LogNormal(log 20, 1)	overdispersion parameter of the reported Beta-Binomial likelihood

Note. Likelihoods: $y \sim \text{Binomial}(n, \text{logit}^{-1}\eta)$ (superseded layer) and $y \sim \text{BetaBinomial}(n, \mu\phi, (1 - \mu)\phi)$ with $\mu = \text{logit}^{-1}\eta$ (reported layer); linear predictor $\eta = \alpha + \beta_{\text{lat}} z_{\text{lat}} + \sigma_{\text{family}} z_{\text{family}[i]}$. Latitude is |Latitude| z-scored with population SD (ddof=0); it is a composite geographic coordinate, not a causal exposure. Family random intercepts model family-level clustering and are NOT full phylogenetic control. The positional model adds language-level intercepts; the initial–final contrast is a within-language comparison. Priors are read from notebook cells 11 and 70, not inferred from posteriors.

S4. MCMC sampling diagnostics

All reported Beta-Binomial models used NUTS with target acceptance 0.99, maximum tree depth 12, 2000 warm-up and 2500 post-warm-up draws per chain across 4 chains. Table S7 reports convergence for every publication-facing model; all show $\hat{R} \leq 1.01$ and zero divergences.

Table S7: Complete MCMC diagnostics for the reported Beta-Binomial models.

Model	Status	N lang.	N fam.	Chains	Warm-up	Draws/chain	Max $\hat{R}$	Min bulk ESS	Min tail ESS	Div.	Target	Seed
Primary (ASJP)	primary	10,886	646	4	2000	2500	1.00	2,062	4,128	0	0.99	20260711+301
Replication (Lexibank)	replication	5,501	298	4	2000	2500	1.00	2,593	4,694	0	0.99	20260711+305
Within/between-family	robustness	10,886	646	4	2000	2500	1.00	1,357	2,504	0	0.99	20260711+302
Initial-vs-final (positional)	robustness	10,886	646	4	2000	2500	1.00	562	990	0	0.99	20260711+303
Hemisphere	robustness	10,886	646	4	2000	2500	1.01	1,428	3,375	0	0.99	20260711+304
Family + macroarea	robustness	10,626	386	4	2000	2500	1.00	2,992	4,630	0	0.99	20260711+306
Matched, no macroarea	robustness	10,626	386	4	2000	2500	1.00	2,056	3,699	0	0.99	20260711+308
Southern macroarea slopes	robustness	4,075	–	4	2000	2500	1.00	2,218	4,203	0	0.99	20260711+307

Note. Leave-family-out and leave-top-two positional re-fits use the same NUTS settings (target 0.99, max_tree_depth 12, 2000 warm-up, 2500 draws/chain, 4 chains) but their $\hat{R}$ -hat/ESS were not saved to separate diagnostics CSVs; their point estimates appear in Tables S10–S11. Southern macroarea per-macroarea family counts are in Table S15. All reported models show 0 divergences.

S5. Binomial versus Beta-Binomial sensitivity

The Binomial fits are retained only as a superseded sensitivity layer. Table S8 pairs the two likelihoods; Figure S1 compares their posterior predictive behaviour on three summary statistics. The Binomial recovers the overall mean but underestimates between-language dispersion; the Beta-Binomial substantially improves dispersion. Both still underpredict the exact-one boundary mass — the Beta-Binomial does not perfectly fit the full distribution. Because no per-bin Binomial predictive output was stored, Figure S1 compares summary statistics only and does not reconstruct a distribution from summaries.

Table S8: Binomial versus Beta-Binomial coefficient comparison (ASJP).

Quantity	Binomial (superseded)	Beta-Binomial (reported)
Primary latitude	-0.190 [-0.203, -0.177]	-0.151 [-0.195, -0.105]
Initial-position latitude	0.009 [-0.026, 0.044]	0.029 [-0.005, 0.063]
Final-position latitude	-0.257 [-0.291, -0.222]	-0.285 [-0.316, -0.252]
Final-minus-initial	-0.266 [-0.275, -0.256]	-0.313 [-0.342, -0.284]
Between-family latitude	-0.512 [-0.648, -0.377]	-0.348 [-0.431, -0.266]
Within-family latitude	-0.083 [-0.089, -0.078]	-0.033 [-0.055, -0.011]

Note. Posterior median [95% credible interval]. The Binomial fits are the earlier layer; the Beta-Binomial fits are the reported layer, adopted because the Binomial understated between-language dispersion (Figure S2).

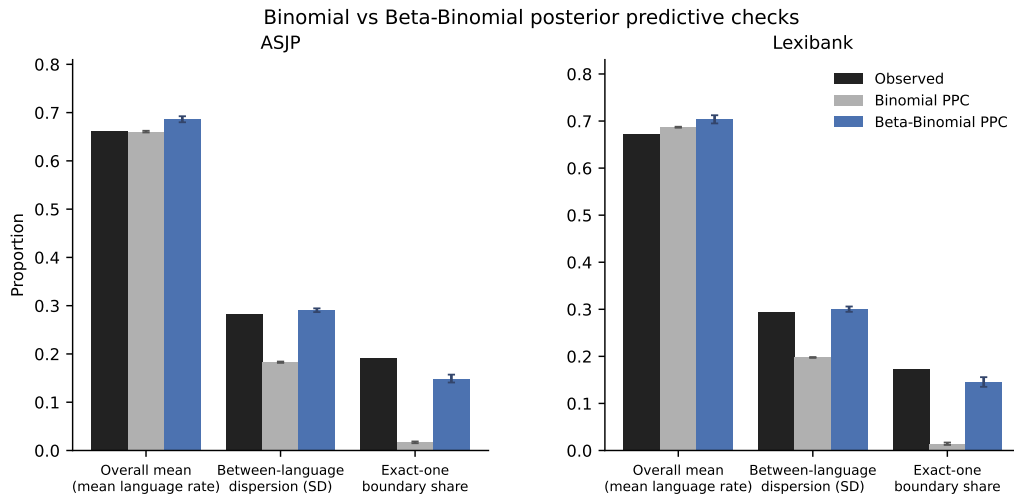


Figure S1: Binomial versus Beta-Binomial posterior predictive comparison for ASJP and Lexibank. Bars show the observed value, the Binomial posterior predictive median, and the Beta-Binomial posterior predictive median (error bars: 95% predictive interval) for the overall mean language rate, the between-language dispersion (standard deviation), and the exact-one boundary share. The overall mean is recovered by both likelihoods; the Binomial underestimates dispersion while the Beta-Binomial recovers it; both underpredict the exact-one boundary share, the Beta-Binomial less severely. Source: `figures/figureS2_source.csv`.

S6. Lexical-coverage robustness

The upper-boundary concentration is not merely a sparse-wordlist artifact. The robustness sweep in Table S9 and Figure S2 varies the minimum *total word count* (not the concept count); the inferential inclusion criterion throughout the paper is a fixed minimum of 20 concepts. Boundary mass consistent with boundary inflation or unresolved mixture structure persists across thresholds. This does not identify a typological class.

Table S9: Boundary mass by minimum total word count.

Database	Min. total words	N lang.	Median total words	Share = 1.00	Share $\geq$ 0.99	Share $\geq$ 0.95
ASJP	20	10,886	40	0.192	0.195	0.257
ASJP	40	5,655	57	0.180	0.184	0.253
ASJP	60	2,699	88	0.179	0.188	0.272
ASJP	80	1,850	96	0.194	0.206	0.292
ASJP	100	743	109	0.106	0.137	0.190
Lexibank	20	5,501	214	0.173	0.237	0.323
Lexibank	40	5,501	214	0.173	0.237	0.323
Lexibank	60	5,501	214	0.173	0.237	0.323
Lexibank	80	5,501	214	0.173	0.237	0.323
Lexibank	100	5,009	230	0.160	0.230	0.311

Note. The robustness sweep varies the minimum *total word count* (`min_n_total`); it is NOT a concept-count sweep. The inferential inclusion criterion throughout the paper is a fixed minimum of 20 concepts. The committed CSV records median total words, not median concepts. Upper-boundary concentration persists across thresholds, so it is not merely a sparse-wordlist artifact; this does not identify a typological class.

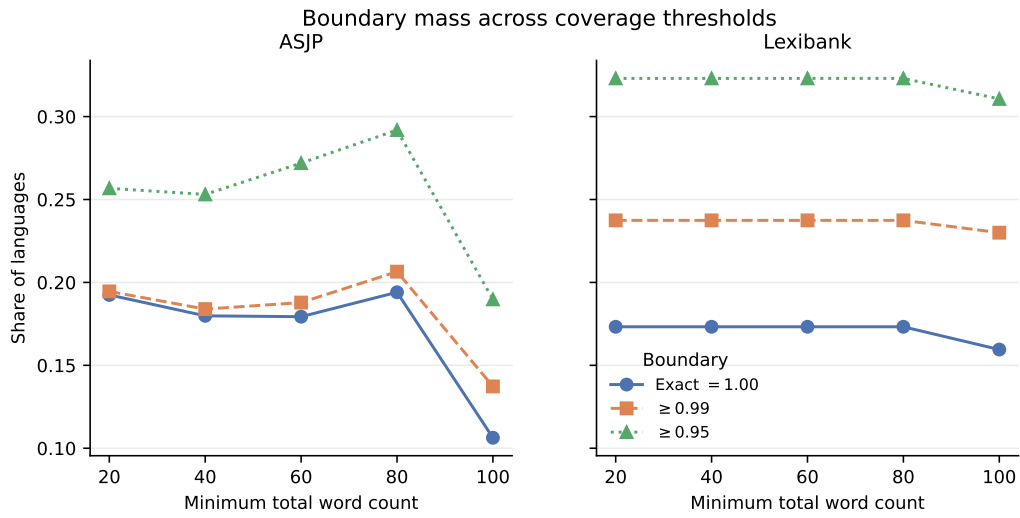


Figure S2: Boundary mass across coverage thresholds for ASJP and Lexibank. Each panel plots the share of languages with word-final vowel rate exactly 1.00, at least 0.99, and at least 0.95, against the minimum total word count. The inferential inclusion criterion is a fixed minimum of 20 concepts; this figure varies only the minimum total word count. Source: `figures/figureS3_source.csv`.

S7. Influence of major language families

The global latitude coefficient is a composite summary. Atlantic-Congo strengthens the negative global slope, whereas Austronesian partly counteracts or suppresses it; the two largest families do not contribute in the same direction (Table S10). Removing both does not eliminate final-position specificity: the final-minus-initial contrast remains clearly negative (Table S11). The association is not genealogy-free.

Table S10: Leave-family-out primary estimates (Beta-Binomial).

Sample	N lang.	N fam.	β_{lat} [95% CrI]	$P(\beta < 0)$
Full primary	10,886	646	-0.151 [-0.195, -0.105]	>0.999
Excl. Atlantic-Congo	8,506	645	-0.069 [-0.119, -0.021]	0.997
Excl. Austronesian	9,353	645	-0.263 [-0.312, -0.214]	>0.999
Excl. both	6,973	644	-0.187 [-0.240, -0.135]	>0.999

Note. Atlantic-Congo strengthens the negative global slope; Austronesian partly suppresses it. The two largest families do not contribute in the same direction, so the global latitude coefficient is a composite summary. The association is not genealogy-free.

Table S11: Positional estimates after excluding Atlantic-Congo and Austronesian.

Parameter	Full [95% CrI]	$P(\beta < 0)$	Excl. both [95% CrI]	$P(\beta < 0)$
Initial slope	0.029 [-0.005, 0.063]	0.048	0.026 [-0.017, 0.068]	0.115
Final slope	-0.285 [-0.316, -0.252]	>0.999	-0.216 [-0.255, -0.177]	>0.999
Final-minus-initial	-0.313 [-0.342, -0.284]	>0.999	-0.242 [-0.276, -0.208]	>0.999

Note. Removing both large families does not eliminate final-position specificity: the final-minus-initial contrast remains clearly negative (-0.242 $[-0.276, -0.208]$). Both intercept and language random effects are included.

S8. Family-level boundary concentration and PPC misfit

Boundary concentration and model misfit are distinct. Atlantic-Congo dominates the exact-one boundary in both databases (Table S12), but its misfit is database-dependent; Austronesian and Nuclear Trans New Guinea show reproducible underprediction across both databases (Table S13 and Figure S3). An exact-one excess is not proof of a discrete open-syllable class. The ASJP **Bookkeeping** label is a Glottolog non-genealogical node; it is excluded from the interpretive family tables and figures and retained in the audit ledger (`audit_asjp_bookkeeping_family.csv`).

Table S12: Largest contributors to exact-one boundary mass.

Database	Family	N lang.	N exact-one	Obs. exact-one share	Share of all exact-one
ASJP	Atlantic-Congo	2,380	931	0.391	0.444
ASJP	Austronesian	1,533	350	0.228	0.167
ASJP	Nuclear Trans New Guinea	408	65	0.159	0.031
ASJP	Sino-Tibetan	995	62	0.062	0.030
ASJP	Central Sudanic	125	56	0.448	0.027
ASJP	Mande	136	47	0.346	0.022
ASJP	Pama-Nyungan	212	47	0.222	0.022
ASJP	Otomanguean	131	42	0.321	0.020
Lexibank	Atlantic-Congo	642	304	0.474	0.319
Lexibank	Austronesian	978	227	0.232	0.238
Lexibank	Otomanguean	88	73	0.830	0.077
Lexibank	Tucanoan	75	49	0.653	0.051
Lexibank	Arawakan	129	43	0.333	0.045
Lexibank	Sino-Tibetan	494	37	0.075	0.039

Note. Atlantic-Congo dominates the exact-one boundary in both databases. The ASJP **Bookkeeping** pseudo-family (17 languages, a Glottolog non-genealogical node) is excluded from this interpretive table but retained in `audit_asjp_bookkeeping_family.csv`. Boundary concentration is descriptive and is not proof of a discrete open-syllable class.

Table S13: Family-level posterior predictive residuals for exact-one boundary mass.

Database	Family	N lang.	Obs. share	PPC median	PPC 95%	Obs – PPC	$p_{\geq \text{obs}}$
ASJP	Atlantic-Congo	2,380	0.391	0.295	[0.270, 0.320]	0.096	0.000
ASJP	Austronesian	1,533	0.228	0.100	[0.082, 0.119]	0.129	0.000
ASJP	Sino-Tibetan	995	0.062	0.054	[0.040, 0.072]	0.008	0.198
ASJP	Indo-European	576	0.014	0.014	[0.005, 0.024]	0.000	0.521
ASJP	Afro-Asiatic	488	0.035	0.025	[0.012, 0.043]	0.010	0.151
ASJP	Nuclear Trans New Guinea	408	0.159	0.071	[0.044, 0.103]	0.088	0.000
ASJP	Tai-Kadai	224	0.000	0.009	[0.000, 0.027]	-0.009	1.000
ASJP	Pama-Nyungan	212	0.222	0.160	[0.104, 0.231]	0.061	0.045
Lexibank	Austronesian	978	0.232	0.088	[0.067, 0.111]	0.144	0.000
Lexibank	Atlantic-Congo	642	0.474	0.458	[0.407, 0.512]	0.016	0.303
Lexibank	Sino-Tibetan	494	0.075	0.043	[0.024, 0.067]	0.032	0.005
Lexibank	Nuclear Trans New Guinea	380	0.068	0.018	[0.005, 0.037]	0.050	0.000
Lexibank	Indo-European	304	0.020	0.003	[0.000, 0.013]	0.016	0.001
Lexibank	Pama-Nyungan	191	0.068	0.068	[0.031, 0.120]	0.000	0.580

Note. Major families ($n \geq 100$ languages) shown; the **Bookkeeping** pseudo-family is excluded (retained in the audit ledger). Boundary concentration and model misfit are distinct: Atlantic-Congo dominates the boundary in both databases but its misfit is database-dependent, whereas Austronesian and Nuclear Trans New Guinea show reproducible underprediction across both. An exact-one excess is not proof of a discrete open-syllable class.

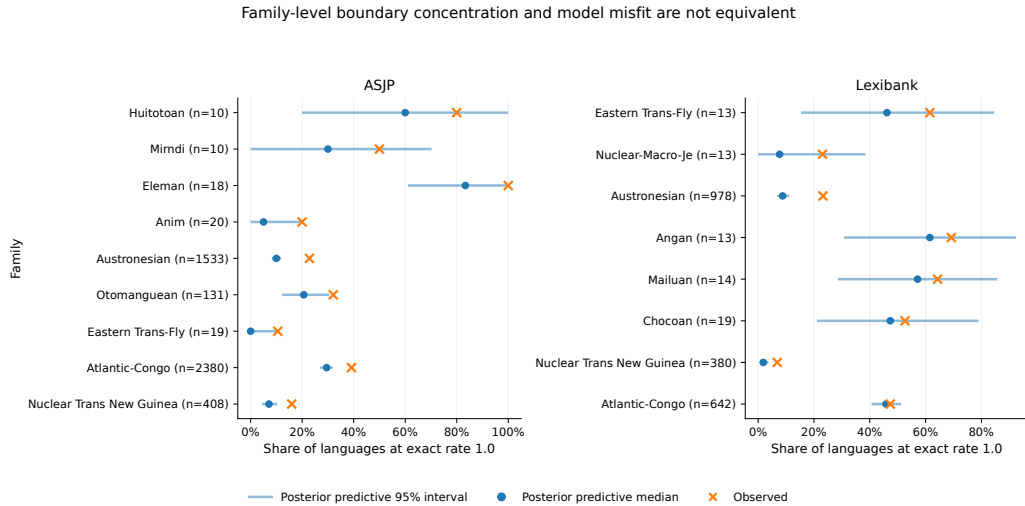


Figure S3: Family-level boundary concentration and model misfit are not equivalent. Existing final figure, reused verbatim. Source: `results/figureS1_family_boundary_ppc.csv`.

S9. Areal and hemispheric heterogeneity

Broad macroarea adjustment removes approximately 48.0% of the matched-sample association (median-based), whereas sample restriction alone produces only a modest change (Table S14). Macroarea is a broad areal adjustment, not a complete areal control. The pooled Southern coefficient is not a universal reversal: the South is regionally heterogeneous (Table S15), and Southern Eurasia ($n = 1$) is uninformative and is not interpreted. Latitude must not be treated as a globally invariant exposure.

Table S14: Matched-sample macroarea adjustment.

Model	N lang.	N fam.	β_{lat} [95% CrI]	OR per SD	Attenuation
Full primary, family-only	10,886	646	-0.151 [-0.195, -0.105]	0.860	–
Matched, family-only	10,626	386	-0.138 [-0.183, -0.092]	0.871	–
Matched, family + macroarea	10,626	386	-0.072 [-0.123, -0.021]	0.931	48.0%

Note. Broad macroarea adjustment removes approximately 48.0% of the matched-sample association (median-based, $1 - 0.0717/0.1377$). Sample restriction alone (full $\rightarrow$ matched) produces only a modest change. Macroarea is a broad areal adjustment, not a complete areal control; latitude is not a globally invariant exposure.

Table S15: Southern-Hemisphere macroarea-specific slopes.

Macroarea	N lang.	N fam.	β [95% CrI]	$P(\beta < 0)$	$P(\beta > 0)$	Status
Papunesia	2,462	124	0.545 [0.444, 0.649]	<0.001	>0.999	informative
Africa	952	12	0.241 [0.114, 0.373]	<0.001	>0.999	informative
South America	354	68	-0.119 [-0.261, 0.021]	0.951	0.049	informative
Australia	306	29	0.239 [0.052, 0.429]	0.006	0.994	informative
Eurasia	1	1	0.204 [-1.156, 1.580]	0.390	0.610	uninformative ($n = 1$)

Note. The pooled Southern coefficient is not a universal reversal: the South is regionally heterogeneous (e.g. South America trends negative, Papunesia strongly positive). Southern Eurasia ($n = 1$) is uninformative and must not be interpreted.

S10. ASJP–Lexibank overlap and comparability

ASJP and Lexibank are independently assembled databases with partially overlapping language coverage; they are not statistically independent language samples. A descriptive overlap audit would require a stable shared identifier. The committed Result CSVs contain no shared cross-database identifier (no Glottocode, and no per-language Lexibank table), and the raw CLDF databases are not part of this repository, so the overlap cannot be computed from the committed evidence package. Table S16 reports the available counts and marks the non-computable cells transparently rather than populating them with fabricated values.

Table S16: Cross-database language overlap (audit; not computable from committed artifacts).

Quantity	Value
ASJP primary languages	10,886
Lexibank languages	5,501
Shared identifiers	not computable
ASJP-only	not computable
Lexibank-only	not computable
Percentage of ASJP overlapping	not computable
Percentage of Lexibank overlapping	not computable
Identifier used	none available (no Glottocode / shared key in committed CSVs)
Unmatched or ambiguous records	not computable

Note. The committed Result CSVs contain no shared cross-database identifier (no Glottocode, and no per-language Lexibank table), and the raw CLDF databases are not in this repository, so the overlap cannot be computed from the committed evidence package.

ASJP and Lexibank are independently assembled databases with partially overlapping language coverage; they are not statistically independent language samples. This cell is marked incomplete rather than populated with fabricated values.

S11. Computational reproducibility

Table S17 records the computational environment, distinguishing exact runtime versions recorded by the notebook from pip installation constraints. Table S18 maps each analysis to its repository and data SHAs, master seed and offset, Result CSV, NetCDF artifact, figure source, and notebook cell. The H100/GPU posterior models were not rerun to build this SI.

Table S17: Computational environment.

Component	Recorded runtime version	pip install constraint	Source
Python	not recorded at runtime	–	Colab runtime
JAX	0.7.2	–	cell 3 stdout
NumPyro	0.19.0	$\geq 0.16, < 0.20$	cell 3 stdout / cell 2 pin
ArviZ	not recorded	$\geq 0.20, < 0.23$	cell 2 pin
pandas	not recorded	$\geq 2.2, < 3$	cell 2 pin
NumPy	not recorded	–	bundled with runtime
SciPy	not recorded	$\geq 1.13, < 2$	cell 2 pin
matplotlib	not recorded	$\geq 3.9, < 4$	cell 2 pin
pyarrow	not recorded	$\geq 17, < 22$	cell 2 pin
netCDF4	not recorded	≥ 1.7	cell 2 pin
Accelerator	CudaDevice(id=0), A100 (declared)	–	cell 3 / notebook metadata
Backend / device	gpu / cuda:0	–	reproducibility_metadata.json

Note. pip constraints are installation bounds, not the exact installed versions. Only JAX and NumPyro versions were printed at runtime; Python, ArviZ, pandas, NumPy and SciPy exact versions were not recorded. `jax_enable_x64=True`; master seed 20260711.

Table S18: Analysis provenance and output-file map.

Analysis	Repo SHA	Data SHA	Master seed	Offset	Result CSV	NetCDF artifact	Figure source	Cell
ASJP H1B primary	d3eff7a0246	0127953496	20260711	+301	asjp_beta_binomial_primary_effect.csv	asjp_beta_binomial_primary.nc	figure2_core_posterior_effects.csv	72
Lexibank H1B replication	d3eff7a0246	edf5f028da	20260711	+305	lexibank_beta_binomial_replication_effect.csv	lexibank_beta_binomial_replication.nc	figure2_core_posterior_effects.csv	80
ASJP H1B within/between	d3eff7a0246	0127953496	20260711	+302	asjp_beta_binomial_within_between_effects.csv	asjp_beta_binomial_within_between.nc	figure2_core_posterior_effects.csv	74
ASJP H1B positional	d3eff7a0246	0127953496	20260711	+303	asjp_beta_binomial_position_effects.csv	asjp_beta_binomial_position_interaction.nc	figure1_position_specific_gradients.csv	76
ASJP H1B hemisphere	d3eff7a0246	0127953496	20260711	+304	asjp_beta_binomial_hemisphere_effects.csv	asjp_beta_binomial_hemisphere.nc	figure2_core_posterior_effects.csv	78
ASJP H1B family macroarea	d3eff7a0246	0127953496	20260711	+306	asjp_beta_binomial_macroarea_effect.csv	asjp_beta_binomial_macroarea_control.nc	figure3_macroarea_attenuation.csv	85
ASJP H1B Southern macroarea	d3eff7a0246	0127953496	20260711	+307	asjp_beta_binomial_south_macroarea_effects.csv	asjp_beta_binomial_south_macroarea_slopes.nc	figure3_southern_macroarea_slopes.csv	86
ASJP H1B matched, no macroarea	d3eff7a0246	0127953496	20260711	+308	asjp_beta_binomial_macroarea_matched_comparison.csv	asjp_beta_binomial_macroarea_matched_no_macroarea.nc	figure3_macroarea_attenuation.csv	100
Leave-family-out	d3eff7a0246	0127953496	20260711	+410	asjp_boundary_family_exclusion_sensitivity.csv	(not committed)	figure2_core_posterior_effects.csv	95
Leave-top-two positional	d3eff7a0246	0127953496	20260711	+420	asjp_position_excluding_top_two_boundary_families.csv	(not committed)	figure2_core_posterior_effects.csv	97

Note. Repo SHA is the results-provenance SHA from `reproducibility_metadata.json`; the SI was built at HEAD 562dee88b5. NetCDF posterior artifacts are named by the notebook but are not committed to the repository; the H100/GPU models were NOT rerun to build this SI.

Data and code availability

All tables and figures in this SI are generated from the Result CSV files in `results/` by the scripts in `supplement/scripts/`. Provenance and publication-facing values are recorded in `supplement/manifest.json`; the per-cell notebook inventory, source ledger, claim-to-evidence map, manuscript contract, and validation/consistency reports are in `supplement/generated/`. Data sources: ASJP [1], Lexibank [2], Glottolog [3]. Inference used NumPyro [4] on JAX [5] with ArviZ [6] diagnostics.

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